



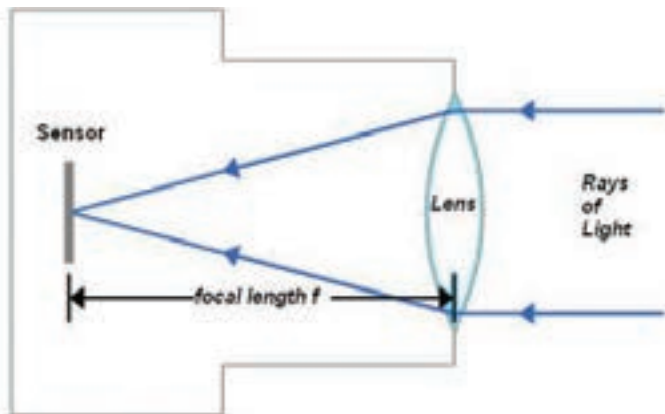
The 'leaf-shutter' kind of diaphragm is made of a number of thin overlapping blades

zines by the character 'f' followed by a number such as $f/1.4$, $f/2$, $f/2.8$ and so on. These are basically gradations that let you adjust the quantity of light or exposure for a particular shot. As you vary the f-stop values, you may notice an apparatus called the 'diaphragm' constricting or expanding behind the lens to permit a correspond-

ing quantity of light.

Technically, an f-stop number is the focal length of the lens divided by the diameter of the aperture. Simply put, the f-number is the ratio of the size of the aperture that the light passes through, to the focal point of the lens.

Each f-number represents a halving of the light intensity from the previous stop and a decrease of the aperture diameter by a factor of $\sqrt{2}$ (around 1.414). Therefore, $f/11$ is half as much as $f/8$, and $f/5.6$ is twice as much light as $f/8$. Currently, most lenses use a standardised f-stop scale – which is a geometric sequence of numbers that correspond to the sequence of the powers of $\sqrt{2}$ – viz. $f/1$, $f/1.4$, $f/2$, $f/2.8$, $f/4$, $f/5.6$, $f/8$, $f/11$,



High school physics now makes sense